CASE 03 · HEALTHCARE · THE FOLLOW-UP BLIND SPOT

LoopBack

Every “come back” said in the OPD becomes a dated, consented follow-up with a named owner — captured at the desk while the patient is still there.

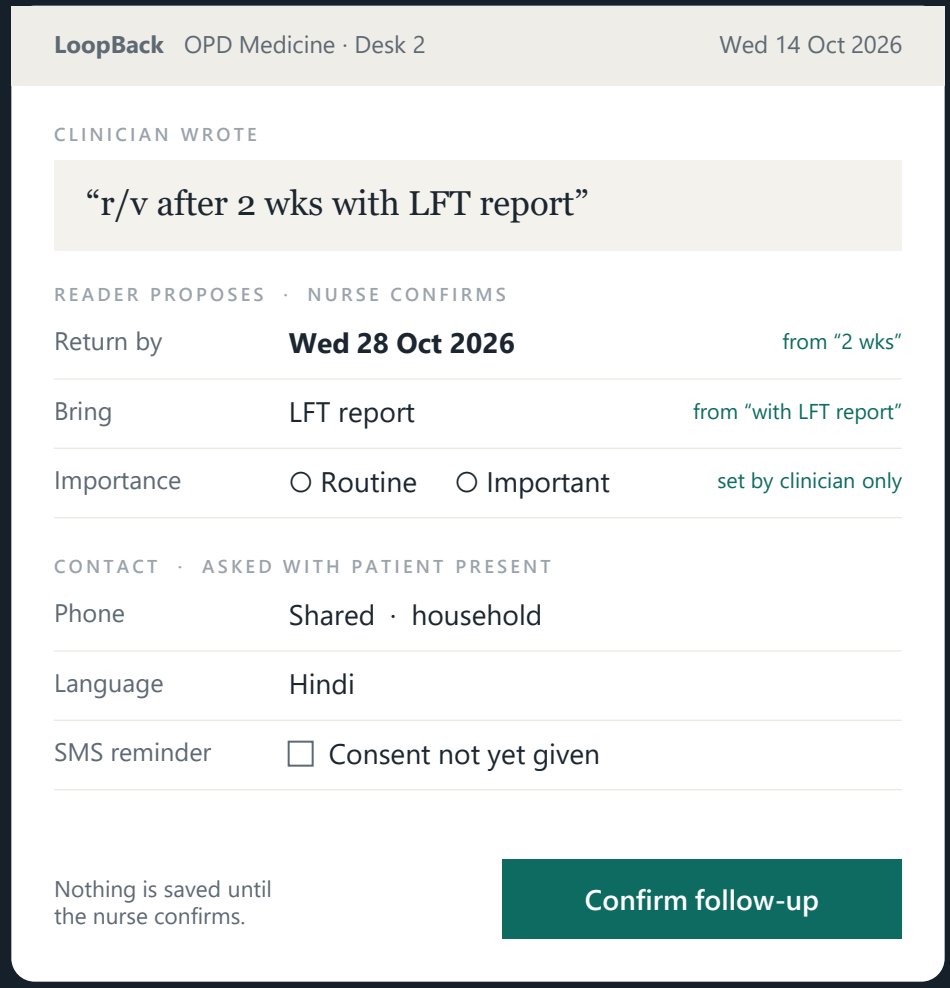
28% of patients marked for follow-up did not return within the recommended window.

CASE competition case file, simulated clinic

TEAM

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Wireframe of the proposed V1 capture screen · illustrative, simulated patient

The follow-up request is lost at the moment it is made

IN THE CLINIC

Clinician says “come back”

Written as free text; wording and detail vary by clinician.

CASE Free-text instructions

→

AT THE DESK

Patient leaves

No due date or bring-list the hospital can track. Some patients travel long distances.

CASE Long travel distances

→

IN THE RECORDS

Nothing to act on

Phone is shared or dead for a quarter of records. No owner, no due date.

CASE 19% shared · 7% unreachable

→

AFTER THE WINDOW

The miss is invisible

28% do not return in time. A small records team cannot tell whom to chase first.

CASE 28% missed the window

Root cause we target: no structured, owned record is created in the first two steps. Reminders, recovery and measurement all depend on that record existing.

WHO IT IS FOR

Primary: patients and caregivers

Daily users: OPD desk nurse, records staff

Also involved: clinicians, ASHA workers, administration

HARD LIMITS (FROM THE CASE) CASE

No diagnosis, treatment change or emergency triage

Human clinical review stays authoritative

No contact on a route the patient has not agreed to

WHAT WE DON’T KNOW YET UNKNOWN

Which misses are clinically important

Which contact routes patients will accept

How varied real follow-up notes are

Four framings, tested against the case before choosing one

FRAMING	FIXES THE ROOT CAUSE?	INSIDE THE SAFETY BOUNDARY?	TESTABLE WITH DATA WE HAVE?	VERDICT
Automated reminders SMS before the visit	○ Partly Needs a due date nobody records	● Yes	● Yes	Kept as one part
Predict who will miss Score urgency or risk	● No	● No Urgency scoring is triage	● No No outcome labels	Rejected
Fix contact data Shared and dead numbers	○ Partly Reaches people, misses stay invisible	● Yes	● Yes	Folded into step 2
Capture → track → recover Create the record at the desk	● Yes	● Yes	● Yes Missed / returned already in registration data	Chosen

We chose the framing that removes the dependency the others share: a reminder, a call list or a prediction all need to know **who is due, by when**, and today nobody records that.

IF A DIFFERENT STAKEHOLDER WERE PRIMARY

Records team

A better call list — still blind to who is due.

Administration

Dashboards and prediction — with no outcomes to learn from.

Patients & caregivers

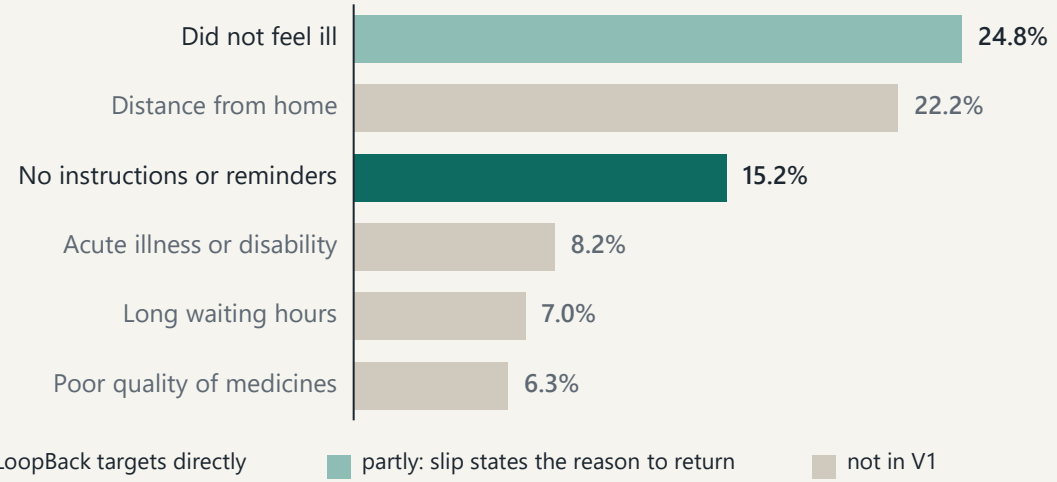
Moves the fix upstream, to the desk, while the patient is present. **Our choice.**

Reminders help — but only once someone knows a visit is due

EXTERNAL

Why patients missed hypertension follow-ups

Public facilities, Bathinda, Punjab · 157 patients who missed · % citing each reason [2]



Takeaway: missing instructions and reminders is a real, top-three reason — but not the only one. Distance stays out of V1, and every closed case records **why** it was missed, so the hospital learns its own pattern.

Different condition and facilities from the case: directional evidence only.

EXTERNAL — Text reminders raised attendance from **67.8% to 78.6%** vs no reminder (RR 1.14, 95% CI 1.03–1.26; 7 RCTs, moderate quality). Trials assumed the patient's own phone. [1]

EXTERNAL — DPDP Act 2023 s.6: consent must be free, specific, informed, unconditional and unambiguous; Rules notified Nov 2025 → **separate consent for each contact route.** [3]

EXTERNAL — Gemini API unpaid tier: inputs may be used to improve products and read by human reviewers → **no identifiers leave the desk.** [4]

CASE — 28% missed the window · 19% shared phone · 7% unreachable · ~900 OPD visits a day · instructions are free text.

ASSUMPTION — If 20–30% of visits need follow-up, a 28% miss rate means **~50–75 new misses a day** — too many to call without a ranked list.

UNKNOWN — Which misses are clinically important; which contact routes patients accept; how varied real notes are.

THE ONE PROBLEM WE CHOSE

An OPD “come back” exists only as free text. The hospital has no record of who is due, by when, why it matters or how to reach them — so the **28% who miss their window** go unnoticed, and a small records team cannot tell whom to chase first, or how.

More important

Every later step — reminders, recovery, measurement — depends on the record that is missing today.

More tractable

Follow-up language is narrow: intervals, dates and a handful of shorthands that rules can read.

More testable

“Missed” and “returned” are already facts in registration data. No new outcome labels needed.

NOT IN V1

Deliberately not solved first: distance and transport, cited by 22% of patients who missed in [2]. V1 only reschedules, and hands cases to an ASHA worker where the patient has agreed.

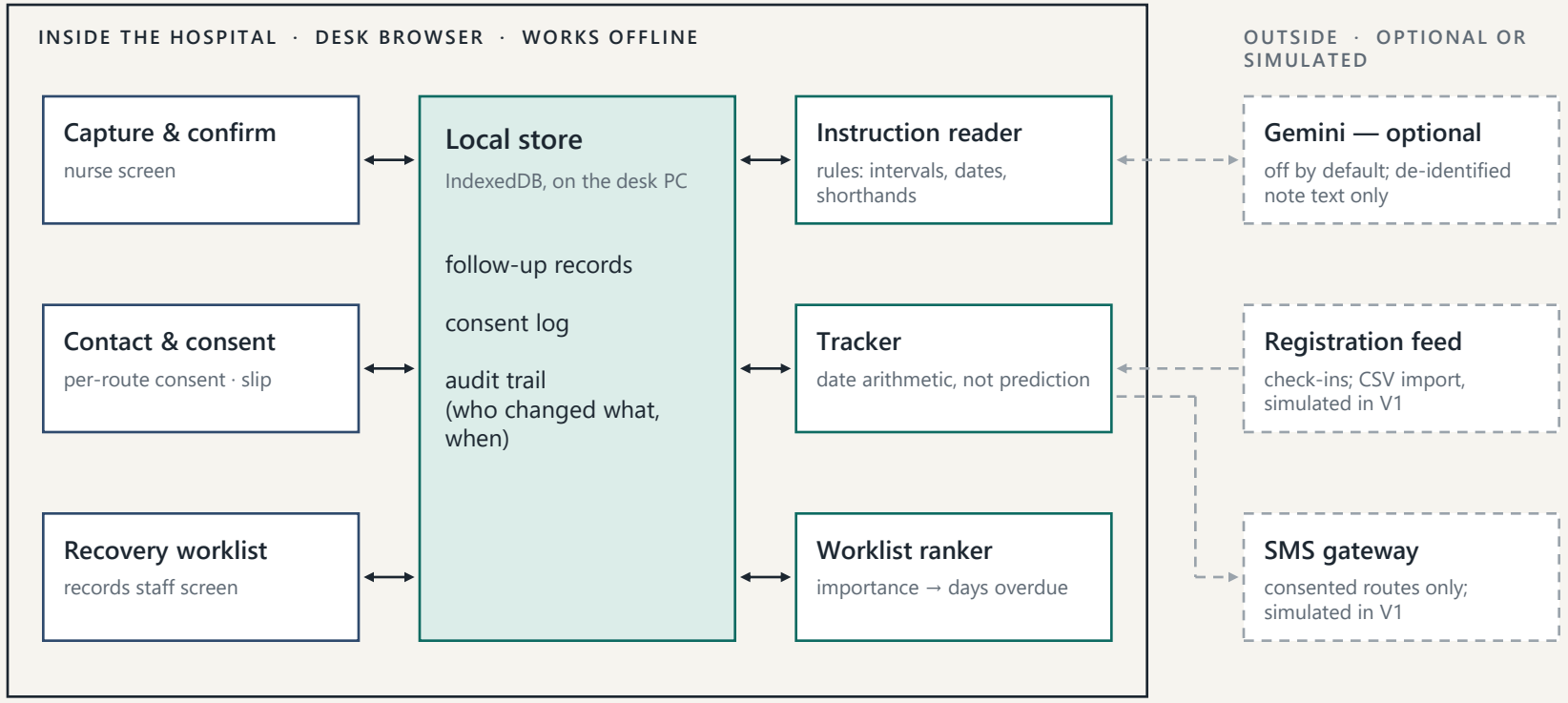
One record, four steps — a person confirms every decision

	1 CAPTURE	2 CHECK CONTACT	3 TRACK	4 RECOVER
CLINICIAN	Writes the instruction as usual. Ticks **Routine** or **Important**.			Reviews any case staff escalate. Clinical judgement stays here.
DESK NURSE	Checks the proposed date and bring-list. **Nothing saves until confirmed.**	Asks whose phone, which language, and **consent for each route** — patient present.		
LOOPBACK system proposes, never decides	Rule reader proposes a date. A vague note is flagged **DECIDE** — it never guesses.	Prints a bilingual slip: date, bring-list, the clinician's reason to return.	Reminder 2 days before, if agreed. Check-in closes the case; none by window end → **Missed**.	Ranks the worklist: importance, then days overdue. **A reason on every row.**
RECORDS STAFF / ASHA			No manual work while the patient is within the window.	Calls on the agreed route. Logs outcome and **miss reason**; unreachable → ASHA, with consent.
	→ Confirmed record	→ Consent log + slip	→ Due / missed / returned	→ Outcome + miss reason

PROPOSED V1

Minimum viable slice for the Build Round: all four steps in one desk browser, offline-first, on simulated patients and a simulated registration feed. Miss-reason codes: felt well · distance/cost · forgot/unclear · seen elsewhere · other.

Offline-first at the desk; identifiers never leave the hospital



Solid: V1 data flow inside the hospital. Dashed: optional or simulated in V1. Teal: system logic. Slate: screens people use.

WHEN INFORMATION IS MISSING OR CONFLICTING

Vague note ("SOS", "review")
No date proposed; flagged DECIDE for the nurse.

Two dates or conflicting notes
Both shown; the nurse picks; the choice is logged.

Shared household phone
Household-safe wording: no clinic, test or diagnosis named.

No working contact route
Alert at the next registration; ASHA handoff only with consent.

Seen elsewhere or moved
Closed only by staff, with a reason code.

The system never diagnoses, triages, changes treatment, gives medical advice, contacts anyone without consent, closes a case on its own, or sends patient identifiers to an AI service.

Tests that can fail, one outcome, one harm metric

ACCEPTANCE TESTS FOR V1

AT1	"r/v 2 wks" becomes a proposed date; nothing saves until the nurse confirms.
AT2	"SOS" or a vague note → no date is invented; the record is flagged DECIDE.
AT3	Importance can only be set by a clinician, never by the system.
AT4	No message names the clinic, test, diagnosis or the clinician's words.
AT5	No reminder goes out on a route without recorded consent.
AT6	A case cannot close without a staff action and a reason code.

V1 intentionally does not predict who will miss, rank by how hard someone is to reach, or send real messages: the SMS gateway is simulated.

BASELINE

CASE

72%

returned on time in the case sample; 28% missed the window.

OUTCOME

Share of follow-ups returned within their window. One OPD, 4 weeks before vs 4 weeks after.

HARM / EXCLUSION

On-time gap between own-phone, shared-phone and unreachable patients. Review the design if the gap exceeds 10 points.

PROCESS

ASSUMPTION

Share of "come backs" captured with a confirmed date; median desk time per capture (target ≤ 90 s).

We claim no improvement figure in advance. In trials, reminders alone moved attendance by ~11 points [1]; our test is whether capture makes that possible here.

Decision trail: where AI changed, sharpened or was overruled

DECISION	INITIAL VIEW	AI CONTRIBUTION	EVIDENCE / VERIFICATION	FINAL HUMAN DECISION
Problem framing	SMS reminder bot	Reminders need a due date; they fail on shared and dead phones	CASE 19% + 7%; Cochrane [1]	**Capture first; reminders kept as one part**
Reading notes	An LLM extracts everything	Flagged invented dates and free-tier data terms	Gemini terms [4]; test notes	**Rule reader first; Gemini optional, de-identified only**
Urgency	AI scores urgency	Proposed a keyword urgency score	Case boundary: no triage	**Rejected — the clinician ticks importance**
Ranking	Rank hardest-to-reach first	Reachability as a rank factor	Fairness check (harm metric)	**Reachability sets the route, never the rank**
Consent	One “agree” box	Summarised DPDP Act s.6	DPDP Act 2023 [3]	**Separate consent per route, nothing pre-ticked**
V1 scope	Reminders fix most misses	Surfaced reason data: felt well 25%, distance 22%, no instructions 15%	Das et al. 2021 [2]	**Added miss-reason codes and reason to return; transport kept out of V1**

Each row links to the genuine AI conversation behind it. **[Add share links before submitting]** — the full decision index is in the submission package.

Every “come back” gets a date, a route and an owner.

V1 WILL

Capture and nurse confirmation
Consent recorded per route
Bilingual slip
Tracking: due, missed, returned
Ranked recovery with miss reasons
Audit trail

V1 WILL NOT

Diagnose, triage or advise
Predict who will miss
Contact without consent
Close a case by itself
Send identifiers to an AI service

LATER — NOT IN V1

Match check-ins via ABHA Scan & Share, live in 8,000+ facilities (Aug 2024) [5]
Real SMS gateway after a consent audit
HMIS integration

DECLARATIONS

All patients and outcomes shown are simulated; case figures are competition material, not real statistics about a named hospital. AI assisted research, drafting and slide design; every consequential decision is in the trail. Reused components: React, Vite, TypeScript, Vitest.

REFERENCES

[1] Gurol-Urganci I, et al. Mobile phone messaging reminders for attendance at healthcare appointments. Cochrane Database Syst Rev 2013;(12):CD007458.

[2] Das B, Neupane D, Gill SS, Singh GB. Factors affecting non-adherence to medical appointments among patients with hypertension at public health facilities in Punjab, India. J Clin Hypertens 2021;23(4):713–719.

[3] Digital Personal Data Protection Act, 2023, s.6; Digital Personal Data Protection Rules, 2025 (notified Nov 2025). MeitY, Government of India.

[4] Google. Gemini API Additional Terms of Service, ai.google.dev/gemini-api/terms (accessed Oct 2026).

[5] MoHFW / National Health Authority press release: 4 crore OPD registrations via ABHA-based Scan & Share, Aug 2024 (mohfw.gov.in/press-info/7476).

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